

# 404-OrbitNotFound

Exploring Space Activity Around Earth

Process Book



Lepère Thomas, Shen Kaifun Kevin, Spiess Alexandre  
EPFL



# The Original Idea



The idea of our project was to create an interactive website that gives users a clear overview of satellites and artificial objects orbiting Earth. We did not aim to build a perfectly accurate physical simulation. Instead, we wanted to make satellite data easier to explore and understand for a non-expert audience.

To do this, we divided the project into three complementary pages. The Orbital Structure page shows where satellites are located around Earth. The Overflights page shows which satellites pass above a selected region. The Space Actors page focuses on who placed objects in orbit, what they sent, when they sent it, and how they compare to others.

Together, these three views turn a technical satellite catalogue into a more accessible and interactive exploration of human activity around Earth.

Early in the project, we defined the three main sections of the website so that each part could focus on a different aspect of satellite activity while still contributing to the same overall exploration.

## OVERVIEW



### ORBITAL STRUCTURE

See where satellites are located around Earth.



### OVERFLIGHTS

Explore which satellites pass above a selected region.



### SPACE ACTORS

Discover who placed objects in orbit, what they sent, when they sent it, and how they compare to others.

## The data

To achieve our objective, we chose to use the SATCAT dataset from CeresTrak. SATCAT is a catalogue of artificial space objects tracked in orbit around Earth. It lists objects such as satellites, space debris, and rocket bodies, and provides basic information for each object, including its name, NORAD catalogue number, launch date, country or organization, object type, operational status, and basic orbital parameters.



## DATA SOURCE

# CeresTrak

Trusted provider of orbital data and satellite tracking information since 1985.

SINCE 1985



# The Landing Page

Because the project was divided into three main sections, we needed a landing page that could introduce the overall topic while guiding users toward the different parts of the website. This page was designed as the first entry point of the project: before entering a specific visualization, users should quickly understand what kind of data we are working with, what questions the website explores, and how the different sections relate to one another.

For this main page, we chose a simple and neutral design that stays close to the visual identity of the rest of the website. The page introduces the project with a short title and description, followed by a few key statistics. These elements give users immediate context without overwhelming them with technical details.

The lower part of the page then acts as a navigation hub, with three cards leading to the main interactive sections: Orbital Structure, Space Actors, and Overflights. Each card briefly summarizes what the section focuses on, so users can choose where to start based on the aspect of satellite activity they are most interested in.

We did not want the landing page to become the main focus of the website or to compete with the visualizations themselves. Instead, its purpose is to provide a clear and efficient transition into the project. It works as a short introduction, giving users enough information to understand the scope of the website before inviting them to explore the three main views in more detail.

SATELLITE DATA VISUALIZATION

## Atlas of Space Activity

Explore how artificial objects populate Earth orbit, who placed them there, and how space activity has evolved over time.

START EXPLORATION

69k+

SATELLITES TRACKED

129+

SPACE ACTORS

3

ORBITAL REGIONS

01

### Orbital Structure

Explore how satellites are distributed across LEO, MEO and GEO.

02

### Space Actors

Compare countries and organizations by payloads, debris and activity over time.

03

### Overflights

Discover which satellites pass above selected regions of the Earth.

# Orbital Structure

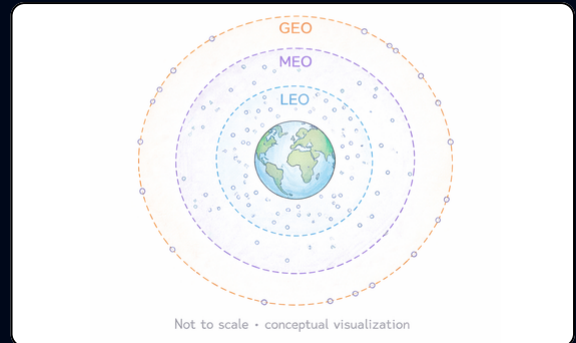


## Concept

The main idea was to represent the distribution of satellites by space actor. Countries and operators cannot officially own regions of space in the same way they can own territory on Earth. However, orbital space is still limited in practice. A satellite physically occupies a position, and safe operation requires a certain amount of separation from other objects. This means that placing many satellites in specific orbital regions can create a form of practical control over those areas.

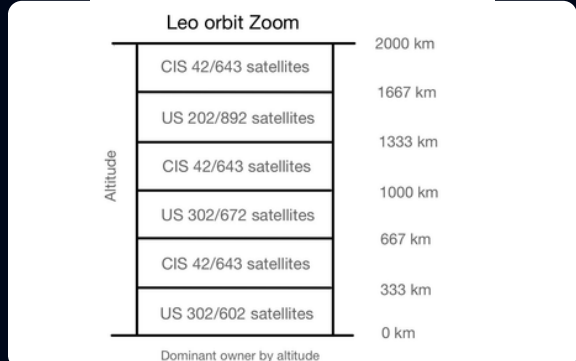
Our first concept was therefore focused on showing which actors were most present in different orbital zones, and how this presence could reveal patterns of influence around Earth.

## Initial designs



## Evolution

The core idea and the data did not change drastically during the design process, because the dataset already supported the story we wanted to tell. What evolved much more was the way users could interact with the visualization. We added ways to zoom into specific orbital regions, access additional explanations, and better understand what was being shown. For example, we introduced explanations for acronyms and controls allowing users to subdivide an orbit into 5, 10, or 15 altitude layers.



## GEO orbit

For LEO and MEO, it made sense to represent satellites through altitude-based layers. But it is not the case for geostationary orbit. GEO satellites appear fixed relative to the Earth's surface, which makes a longitude-based map more meaningful than an altitude-layer representation. This is how the idea for the final GEO plot emerged. This representation also fits better the nature of GEO itself. It is located around 35,786 km above Earth's surface and has an inclination close to 0°. So it is essentially organized along a narrow equatorial belt making the available positions very limited.

As a result, each satellite placed in GEO has a particularly high strategic value. Orbital slots, international coordination, unauthorized maneuvers, and competition for useful longitudes can all become sources of tension between countries and operators. GEO is also harder to access and maintain than LEO or MEO, which makes every operational satellite in this orbit even more significant.



# Overflights



## Concept

The overflight visualization was designed to show how satellites move above specific countries over a limited period of time. While the orbital structure visualization focuses on the spatial distribution of satellites around Earth, this view introduces a more dynamic perspective. It allows users to observe which satellites pass over a selected country and how frequently these passages occur within a given time window.

## Evolution

The core idea of the overflight visualization also remained relatively stable during the design process. What changed most was the organization of the interface. The final version places the time slider more clearly at the top and groups satellites by owner in a dedicated side panel.

## Why Overflights Matter

They make the relationship between satellites and territories more concrete. A satellite in orbit may seem distant and abstract, but its ground track shows that it repeatedly passes over specific regions of Earth. This creates a form of visibility from space: depending on its mission, a satellite can potentially observe, communicate with, or collect data from areas located below its path. Looking at overflights therefore helps connect orbital activity to geographic and political realities on the ground. It also makes it easier to compare how different space actors are present above different countries, not only through the number of satellites they operate, but through the frequency and distribution of their passages.

This approximation is not meant to provide a legal or perfectly exact definition of overflight. Instead, it offers a practical and visually effective way to highlight patterns of satellite presence above different parts of the world. The goal of this view is therefore not to produce an exact surveillance map, but to help users understand how often satellites from different actors move over selected countries, and how orbital motion creates repeated forms of visibility and presence from space.

## Initial Design



## Final Version



# Space Actors

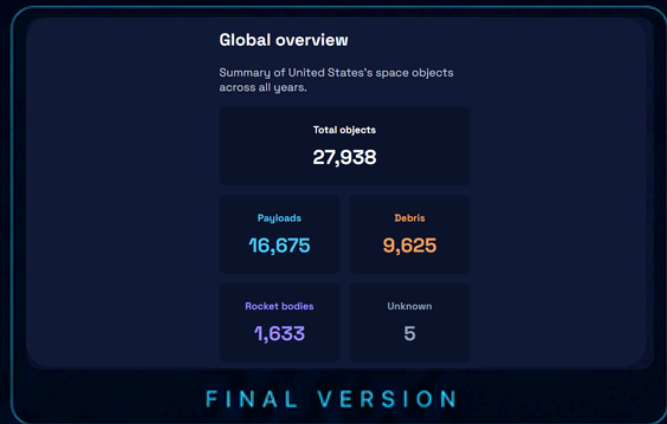
The initial idea of this page was to help users understand which space actors placed objects in orbit, what kind of objects they contributed, when this activity happened, and how each actor compares to the others.

The final page is composed of four main visual components: Global Overview, Temporal Activity, World Ranking, and Top Actors Over Time.



## GLOBAL OVERVIEW

The first component is the Global Overview. The idea is to summarize the selected actor with KPI cards showing total objects, payloads, debris, rocket bodies and unknown objects. This gives the user a quick understanding of the actor's overall footprint.



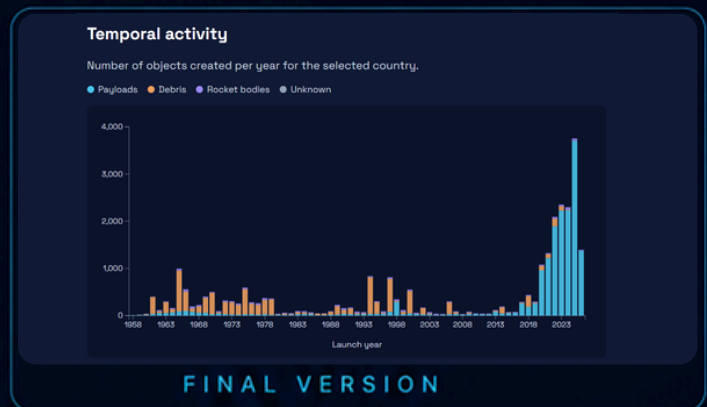
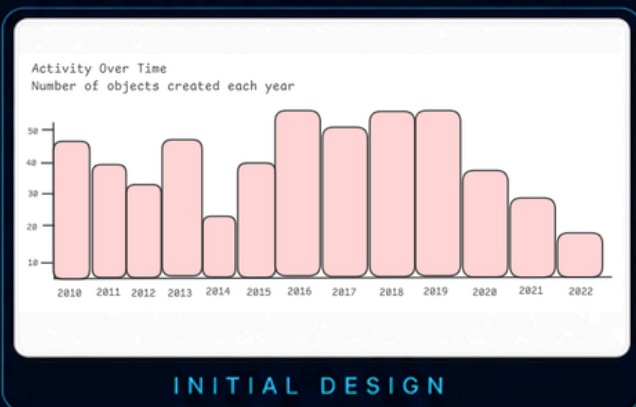
The central idea remained the same: giving a quick summary of the selected actor. However we added consistent colors for each object category: payloads, debris, rocket bodies and unknown objects.



## TEMPORAL ACTIVITY

The second component is the Temporal Activity chart. It shows when the selected actor contributed objects to orbit.

### THE EVOLUTION



In the initial design, we planned a simple yearly bar chart. This was useful to show when activity happened, but it only showed the total number of objects per year. In the final version, we changed it into a stacked bar chart so that users can see both the number of objects launched each year and the composition of these objects by type.



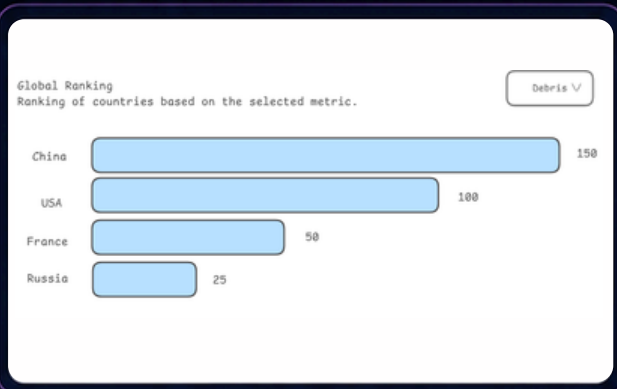
This part focuses on comparison between actors. These views help users understand how actors rank against each other and how the dominant contributors have changed over time.



## WORLD RANKING

The third component is the World Ranking. It compares all space actors according to a selected metric, such as total objects, payloads, debris or rocket bodies.

### THE EVOLUTION



#### INITIAL DESIGN



#### FINAL VERSION

The initial idea already included a horizontal ranking chart with a metric selector. In the final version, we kept this concept but made it clearer and more interactive. The selected actor is highlighted in orange, and its current rank is shown in the subtitle. We also added smooth transitions when the metric changes, which makes the ranking easier to follow.



## TOP ACTORS OVER TIME

The fourth component is Top Actors Over Time.



#### FINAL VERSION

It was added later in the design process to provide a broader historical comparison between the largest actors.

While the Temporal Activity chart focuses on one selected actor, this line chart compares the evolution of the top actors over time. It helps users understand how space activity shifted historically between major actors.



# Data Challenges

One of the main challenges came from the structure of the SATCAT dataset. SATCAT uses short codes to identify the owners of space objects, such as US, PRC, CIS, or EUME. These codes are useful in the raw dataset, but they are not always understandable for users. To make the visualizations clearer, we created a shared mapping file that converts these codes into readable names and ensures consistent naming across the different pages of the website.

Another issue was that some official owner names were too long. For example, European Organization for the Exploitation of Meteorological Satellites (EUMETSAT) is too long to display comfortably in charts, dropdown menus, and ranking labels. We therefore manually shortened some names in the owner mapping file, while keeping them recognizable for users.

A further filtering step was needed for the orbital data. The dataset includes objects that are not necessarily active satellites orbiting Earth, or whose orbital information was not relevant to our visualizations. To avoid plotting misleading points, we removed entries that did not correspond to satellites in Earth orbit.

For the overflight visualization, our first idea was to let users explore satellite trajectories over a period of up to 48 hours. However, this quickly made the map too cluttered. Displaying too many trajectories at once reduced readability and made it harder to identify meaningful patterns. For this reason, the maximum time window was reduced to 24 hours, which still shows several satellite passages while keeping the map visually manageable.

Another challenge was the large number of satellites passing over some countries. Showing all of them individually would have made the interface difficult to navigate, especially for countries crossed by many orbital paths. To solve this, satellites were grouped by space actor. This made the list easier to read and also reinforced the geopolitical angle of the project by showing which actors were most represented in the overflights.

The main technical difficulty was computing satellite trajectories and detecting whether they passed over a country. Since using exact country borders would have been more complex and computationally heavier, we used a simplified rectangular area matching each country's approximate extent. If a satellite trajectory intersected this area during the selected time window, it was considered an overflight.





# Team contribution

The project was divided into three complementary parts, with each team member mainly responsible for one visualization page. At the same time, we collaborated on the overall website structure, visual consistency, integration, debugging, final polishing and the process book.

**Lepère Thomas** was responsible for the **Overflights** page, for the preliminary data analysis in Milestone 1 and the screencast.

**Shen Kaifun Kevin** was responsible for the **Orbital Structure** page, and also helped with the overflights page, the main page, the design and the writing of the process book.

**Spiess Alexandre** was responsible for the **Space Actors** page, for the writing of the process book and contributed on the main page.

# Links

## Website:

<https://com-480-data-visualization.github.io/404-OrbitNotFound/>

## GitHub Repository:

<https://github.com/com-480-data-visualization/404-OrbitNotFound>

## SATCAT Dataset:

<https://celestrak.org/satcat/>